Quantum Noise Simulation

From density matrices and Kraus operators to noisy circuits in CUDA-Q — explained step by step

CUDA-Q Quantum Algorithm Benchmarking Suite

Companion lecture to `Noise_Simulation.ipynb`

All results in these slides were computed with NVIDIA CUDA-Q 0.16.0 on a CPU (`stim`, `density-matrix-cpu`, `qpp-cpu`; no GPU) and checked against independent NumPy calculations. No physical quantum processor was used.

What you will learn today

1. **Ideal vs noisy** quantum computing: sources of noise, decoherence, fidelity
2. **Density matrices**, pure and mixed states, **quantum channels**, Kraus operators, CPTP maps
3. The standard **noise channels**: bit, phase and bit-phase flip, depolarizing, amplitude and phase damping
4. Hardware noise: two-qubit gates, **readout**, T_1 and T_2, error rates
5. **Simulating noise** with CUDA-Q: the code of `Noise_Simulation.ipynb` line by line, results, benchmarks
6. **Mitigation and correction**: ZNE, readout mitigation, dynamical decoupling, error correction; hardware outlook

You only need

complex numbers, 2×2 matrices, basic probability.

Three kinds of statements

IN REPOSITORY code that exists in this repository

THEORY ONLY explained, not implemented here

ILLUSTRATIVE EXPERIMENT small experiments run for this lecture (`noise_source/make_figures.py`)

What this repository actually contains

Topic	Status	Where
GHZ and mirror circuits; 4 noise models after every gate	IN REPOSITORY	`Noise_Simulation.ipynb`
`stim`, density-matrix and GPU-trajectory methods	IN REPOSITORY	`...` + `common_utils.py`
population, parity, Hellinger, TVD, fidelity, purity, $(1-p)^G$	IN REPOSITORY	`Noise_Simulation.ipynb`
channels on the Bloch sphere, T_1/T_2, coherent and readout errors, ZNE, DD, bit-flip code, noisy QFT/QAOA/VQE, simulation cost	ILLUSTRATIVE EXPERIMENT	`make_figures.py`
readout noise in the notebook, RB, surface codes, hardware runs	THEORY ONLY	—

No GPU, multi-GPU or quantum-hardware run was performed: the trajectory (GPU) path of the notebook exists but was not executed. Qiskit/Aer or other simulators are not used anywhere in the repository.

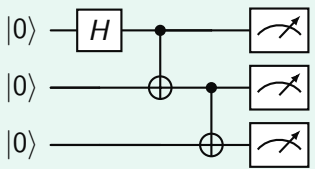
Ideal quantum computing

An ideal quantum computer

- starts in a perfect state $|0\cdots0\rangle$
- applies perfect unitary gates: $|\psi\rangle \mapsto U|\psi\rangle$, $U^\dagger U = I$
- stays in a **pure** state at all times
- measures perfectly: $P(x) = |\langle x|\psi\rangle|^2$ (Born rule)

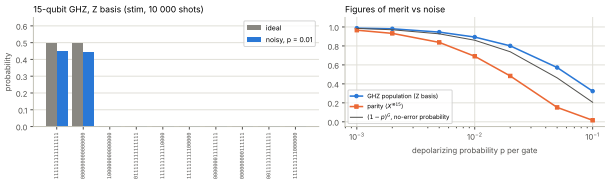
Only **shot noise** remains: probabilities are estimated from a finite number of runs, with error $\approx \sqrt{P(1-P)/S}$.

The GHZ state of the notebook



$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0\cdots0\rangle + |1\cdots1\rangle)$: ideally only two outcomes, 50 % each, and parity $\langle X^{\otimes N}\rangle = 1$.

Ideal versus noisy circuits IN REPOSITORY



Notebook default: 15-qubit GHZ, depolarizing noise after every gate, `stim`, 10 000 shots.

- ideal: 2 outcomes; at $p = 0.01$: 96 outcomes, population 0.8936
- the parity $\langle X^{\otimes 15} \rangle$ falls faster: 0.6910
- at $p = 0.1$: population 0.3232, parity 0.0160, 1165 outcomes

Re-executed for these slides: identical to the recorded outputs.

Sources of quantum noise

source	example	category	typical model
control imperfections	pulse amplitude 1 % too large	coherent error	extra unitary $e^{-i\varepsilon H}$
environment: energy exchange	photon emission, two-level defects	decoherence (T_1)	amplitude damping
environment: fluctuating fields	flux / charge noise, nuclear spins	decoherence (T_2, T_φ)	phase damping, random phases
gate implementation	finite fidelity of two-qubit gates	stochastic gate noise	depolarizing / Pauli channels
crosstalk	neighbour gates rotate idle qubits	correlated coherent error	multi-qubit unitaries
leakage	excitation to $\|2\rangle$ in transmons	outside the qubit space	qutrit models
state preparation and measurement	thermal population, misassigned outcomes	SPAM / readout error	confusion matrix

The notebook models the fourth row (Pauli channels) and amplitude damping. Crosstalk, leakage, drift and readout errors are *not* modelled — a central difference between simulated and real hardware noise.

Decoherence: entanglement with the environment

A qubit interacts with its environment E:

$$(\alpha\,|0\rangle + \beta\,|1\rangle)\,|e\rangle \mapsto \alpha\,|0\rangle\,|e_0\rangle + \beta\,|1\rangle\,|e_1\rangle\,.$$

Tracing out E:

$$\rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^*\,\langle e_1|e_0\rangle \\ \alpha^*\beta\,\langle e_0|e_1\rangle & |\beta|^2 \end{pmatrix}.$$

The coherence shrinks by $|\,\langle e_0|e_1\rangle\,| \le 1$: the environment "learned" which state the qubit is in.

Two worked examples

Entanglement: $\sqrt{0.8}\,|00\rangle + \sqrt{0.2}\,|11\rangle$ gives $\rho_A = \mathrm{diag}(0.8, 0.2)$, purity 0.68, although the global state is pure.

Random phase: $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}\,|1\rangle)$, φ uniform in $[-\frac{\pi}{2}, \frac{\pi}{2}]$: $\overline{\langle X\rangle} = \frac{2}{\pi} = 0.64$.

Decoherence vs. coherent error

Decoherence makes the state *mixed* (information leaks out); a coherent error keeps it pure but wrong.

Quantum state fidelity

- pure vs. pure: $F = |\langle\psi|\phi\rangle|^2$
- pure target, mixed state: $F = \langle\psi|\rho|\psi\rangle$ — used by the notebook
- general: $F(\rho,\sigma) = \left(\operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right)^2$ (some texts omit the square)
- GHZ: $F = \frac{1}{2}(\rho_{0\cdots0,0\cdots0} + \rho_{1\cdots1,1\cdots1}) + \operatorname{Re}\rho_{0\cdots0,1\cdots1}$ — needs the coherence

Numerical examples

$\rho = 0.8\,|+\rangle\langle+| + 0.2\frac{I}{2}$: $F = 0.8 + 0.1 = 0.9$.

Tomography $\rho = \left(\begin{smallmatrix} 0.55 & 0.35 \\ 0.35 & 0.45 \end{smallmatrix}\right)$, target $|+\rangle$: $F = \frac{1+r_x}{2} = 0.85$.

6-qubit GHZ, depolarizing $p = 0.01$ after every gate (density matrix, notebook model): $F = 0.9492$, purity 0.9013.

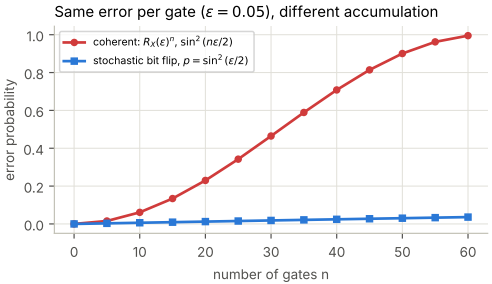
Classical counterpart (from samples): Hellinger fidelity $F_H = (\sum_x \sqrt{p_x q_x})^2$.

Quantum gates and imperfect operations ILLUSTRATIVE EXPERIMENT

Coherent error: X implemented as $R_X(\pi + \varepsilon)$: $P(\text{error}) = \sin^2 \frac{\varepsilon}{2}$ ($\varepsilon = 0.1$: 0.0025).

Repeated n times, the angles *add*: $\sin^2 \frac{n\varepsilon}{2}$ — grows like n^2.

Stochastic error with the same per-gate probability $p = \sin^2 \frac{\varepsilon}{2}$: $\frac{1}{2}[1 - (1-2p)^n] \approx np$ — grows like n.



$\varepsilon = 0.05$, $n = 60$: coherent 0.9950, stochastic 0.0361 (CUDA-Q).

Same average gate fidelity, wildly different circuits: coherent errors are the most dangerous. Pauli twirling (slide "Twirling") converts them into the stochastic Pauli noise that the notebook simulates.

Density matrices

An ensemble "$|\psi_i\rangle$ with probability p_i" is described by

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \, .$$

Properties: $\rho = \rho^\dagger$, $\mathrm{Tr}\,\rho = 1$, $\rho \geq 0$.

- probabilities: $P(k) = \rho_{kk}$; expectation values $\langle M \rangle = \mathrm{Tr}(\rho M)$
- gates: $\rho \mapsto U \rho U^\dagger$
- N qubits: $2^N \times 2^N$, i.e. 4^N numbers ($16 \cdot 4^N$ bytes)

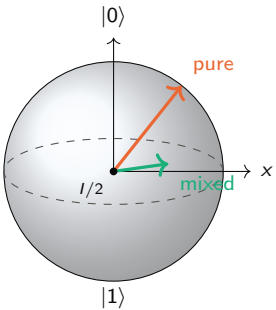
Examples

$\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \mapsto \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$ (pure).

$\begin{pmatrix} 0.6 & 0.5 \\ 0.5 & 0.4 \end{pmatrix}$ is *not* a state: $\det = -0.01 < 0$.

Two independent qubits $\mathrm{diag}(0.9, 0.1) \otimes \mathrm{diag}(0.8, 0.2)$:
$P(\text{equal}) = 0.72 + 0.02 = 0.74$.

Pure and mixed states on the Bloch ball



$\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$, $|\vec{r}| \le 1$.

- pure $\Leftrightarrow$ $|\vec{r}| = 1 \Leftrightarrow \mathrm{Tr}\,\rho^2 = 1$; mixed $\Leftrightarrow$ inside the ball
- $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$; eigenvalues $\frac{1\pm|\vec{r}|}{2}$
- maximally mixed $I/2$ at the centre; N qubits: purity $\ge 2^{-N}$

Worked example

$\rho = \left(\begin{smallmatrix} 0.7 & 0.3 \\ 0.3 & 0.3 \end{smallmatrix}\right)$: $\vec{r} = (0.6, 0, 0.4)$, purity 0.76, eigenvalues 0.861, 0.139. Noise channels act on $\vec{r}$ as **affine maps** that shrink (and may shift) the ball.

Quantum channels and Kraus operators

Every physical noise process on a system is a **quantum channel** with a **Kraus representation**

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I.$$

Interpretation: with probability $p_k = \mathrm{Tr}(K_k \rho K_k^\dagger)$ "event k" happens and the state becomes $K_k \rho K_k^\dagger / p_k$.

- it arises from a unitary on system + environment, followed by tracing out the environment (Stinespring)
- a d-dimensional channel needs at most d^2 Kraus operators

Worked example: completeness

$K_0 = \mathrm{diag}(1, 0.8)$, $K_1 = b\,|0\rangle\langle 1|$:
$K_0^\dagger K_0 + K_1^\dagger K_1 = \mathrm{diag}(1, 0.64 + b^2)$, so $b = 0.6$ — amplitude damping with $\gamma = 0.36$.

Worked example: applying a channel

Bit flip $p = 0.2$ on $\left(\begin{smallmatrix} 0.9 & 0.3 \\ 0.3 & 0.1 \end{smallmatrix}\right)$:
$0.8\rho + 0.2 X\rho X = \left(\begin{smallmatrix} 0.74 & 0.30 \\ 0.30 & 0.26 \end{smallmatrix}\right)$.

Completely positive trace-preserving (CPTP) maps

A map $\mathcal{E}$ is a valid quantum operation iff it is

- **linear**,
- **trace preserving**: $\mathrm{Tr}\,\mathcal{E}(\rho) = \mathrm{Tr}\,\rho \Leftrightarrow \sum K^\dagger K = I$,
- **completely positive**: $(\mathcal{E} \otimes \mathrm{id})(\rho_{AB}) \geq 0$ for every state of system + any ancilla.

Kraus form $\Rightarrow$ CP automatically. The transpose $\rho \mapsto \rho^{\mathsf{T}}$ is positive but *not* CP — not physical.

Choi state: $J(\mathcal{E}) = (\mathcal{E} \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$; $\mathcal{E}$ is CP iff $J \geq 0$; the process fidelity is $\langle\Phi^+|J|\Phi^+\rangle$.

Kraus operators are not unique

$\{\sqrt{1-p}I, \sqrt{p}Z\}$ (phase flip) and $\{\mathrm{diag}(1, \sqrt{1-\lambda}), \mathrm{diag}(0, \sqrt{\lambda})\}$ (phase damping) are the same channel when $1 - 2p = \sqrt{1-\lambda}$, e.g. $\lambda = 0.36 \leftrightarrow p = 0.1$.

Check

$\{I, \sqrt{0.1}X\}$: $\sum K^\dagger K = 1.1I$ — not trace preserving. $\{\sqrt{0.7}I, \sqrt{0.3}Y\}$ — valid (bit-phase flip).

Bit-flip channel

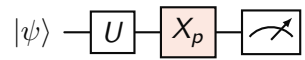
Physics: an unwanted X (e.g. a spurious π pulse) with probability p.
Kraus: $K_0 = \sqrt{1-p}\, I$, $K_1 = \sqrt{p}\, X$.

$$\mathcal{E}(\rho) = (1-p)\rho + pX\rho X,$$

$$(r_x, r_y, r_z) \mapsto (r_x,\ (1-2p)r_y,\ (1-2p)r_z).$$

Effect: the ball shrinks towards the x-axis; $|\pm\rangle$ are immune.
Example: $|0\rangle$, $p = 0.1$: $\rho = \mathrm{diag}(0.9, 0.1)$, $P(1) = 0.1$, purity 0.82.

Circuit level (X_p: X with probability p)



CUDA-Q: `cudaq.BitFlipChannel(p)`. In the notebook (`bit_flip`): after every `h`, `x`, `ry`, `rz`, and independently on both qubits after every CNOT (`Pauli2`: IX, XI with $p(1-p)$, XX with p^2). IN REPOSITORY

Phase-flip channel

Physics: a random phase kick Z (dephasing).
Kraus: $\sqrt{1-p}\,I$, $\sqrt{p}\,Z$.

$$\mathcal{E}(\rho) = \begin{pmatrix} \rho_{00} & (1-2p)\rho_{01} \\ (1-2p)\rho_{10} & \rho_{11} \end{pmatrix}.$$

Effect: $(r_x, r_y, r_z) \mapsto ((1-2p)r_x, (1-2p)r_y, r_z)$; populations unchanged.
Example: $|+\rangle$, $p = 0.1$: $\langle X\rangle = 0.8$, $P(+) = 0.9$; $p = \frac{1}{2}$ destroys all coherence, $p = 1$ is just a Z gate.

Invisible in the Z basis

15-qubit GHZ with phase-flip noise $p = 0.05$ (`stim`): population 1.0000, but X-basis parity 0.0504. One Z error maps $|0\cdots0\rangle + |1\cdots1\rangle \to |0\cdots0\rangle - |1\cdots1\rangle$.

CUDA-Q: `cudaq.PhaseFlipChannel(p)`; notebook: `phase_flip` (`Pauli2`: IZ, ZI, ZZ).
IN REPOSITORY

Bit-phase-flip channel

Physics: both a flip and a phase error, $Y = iXZ$.
Kraus: $\sqrt{1-p}\, I$, $\sqrt{p}\, Y$.

$$(r_x, r_y, r_z) \mapsto ((1-2p)r_x,\ r_y,\ (1-2p)r_z).$$

Effect: shrink towards the y-axis; it flips Z outcomes *and* the sign of $\langle X \rangle$.
Example: $p = 0.3$ on $|0\rangle$: $\langle Z \rangle = 0.4$; on $|+\rangle$: $\langle X \rangle = 0.4$ (measured with CUDA-Q `YError`).

Pauli channels in general

$\mathcal{E}(\rho) = p_I \rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z$ maps $r_k \mapsto (1 - 2(p_j + p_l))r_k$ (j, l the other two Paulis). Bit, phase, bit-phase flip and depolarizing are special cases. THEORY ONLY for the notebook (library: `cudaq.YError`, `Pauli1`).

Depolarizing channel

Physics: an unknown random Pauli error — the standard "average" gate error.
Kraus (CUDA-Q): $\sqrt{1-p}\,I$, $\sqrt{p/3}\,X$, $\sqrt{p/3}\,Y$, $\sqrt{p/3}\,Z$.

$$\begin{aligned}\mathcal{E}(\rho) &= (1-p)\rho + \tfrac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)\\ &= (1-\lambda)\rho + \lambda\tfrac{I}{2}, \qquad \lambda = \tfrac{4p}{3}.\end{aligned}$$

Effect: $\vec{r} \mapsto (1-\frac{4p}{3})\vec{r}$ — isotropic shrink.
Example: $p = 0.03$ on $|0\rangle$: $P(1) = \frac{2p}{3} = 0.02$.
Measured: `DepolarizationChannel(0.3)` on $|0\rangle$ gives $\langle Z\rangle = 0.6$.

Two-qubit version (`Depolarization2`)

$\sqrt{1-p}\,II$ and the 15 other two-qubit Paulis with probability $p/15$ each. Probability that the control's Z outcome flips: $8p/15$ (X or Y on the control).

Conventions differ

Some texts write $(1-p)\rho + pI/2$; then CUDA-Q's p equals $\frac{3}{4}$ of theirs. Always check the Kraus operators.

Default noise of the notebook. IN REPOSITORY

Amplitude damping (energy relaxation, T_1)

Physics: $|1\rangle$ decays to $|0\rangle$ by emitting energy.
Kraus: $K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}$ (no jump),
$K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$ (jump).

$$\mathcal{E}(\rho) = \begin{pmatrix} \rho_{00} + \gamma\rho_{11} & \sqrt{1-\gamma}\,\rho_{01} \\ \sqrt{1-\gamma}\,\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix}.$$

Effect: $r_{x,y} \mapsto \sqrt{1-\gamma}\, r_{x,y}$, $r_z \mapsto \gamma + (1-\gamma) r_z$ — **non-unital**: $I/2$ is not a fixed point.

Worked example

$|+\rangle$, $\gamma = 0.36$: $\begin{pmatrix} 0.68 & 0.40 \\ 0.40 & 0.32 \end{pmatrix}$.
Notebook model on a 6-qubit GHZ, $\gamma = 0.05$ (density matrix): $P(0\cdots0) = 0.5262$, $P(1\cdots1) = 0.2844$ — outcomes biased towards 0; fidelity 0.7824.

CUDA-Q: `AmplitudeDampingChannel`; on CNOTs the notebook uses a 4-operator `KrausChannel` ($K_i \otimes K_j$). Not a Pauli channel: no `stim`. IN REPOSITORY

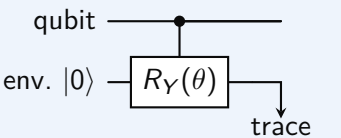
Phase damping (pure dephasing, T_φ)

Physics: the environment learns the energy without taking it; slow phase fluctuations.
Kraus: $K_0 = \mathrm{diag}(1, \sqrt{1-\lambda})$, $K_1 = \mathrm{diag}(0, \sqrt{\lambda})$.

$$\mathcal{E}(\rho) = \begin{pmatrix} \rho_{00} & \sqrt{1-\lambda}\,\rho_{01} \\ \sqrt{1-\lambda}\,\rho_{10} & \rho_{11} \end{pmatrix}.$$

Effect: shrink towards the z-axis; unital.
Example: $|+\rangle$, $\lambda = 0.36$: $\langle X \rangle = 0.8$ — the same as a phase flip with $p = 0.1$.

Circuit picture



controlled rotation of the environment,
$\cos\frac{\theta}{2} = \sqrt{1-\lambda}$

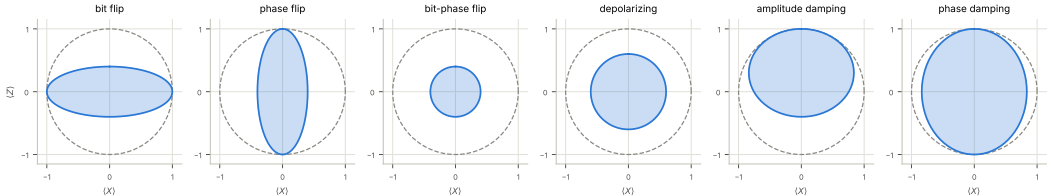
CUDA-Q library: `cudaq.PhaseDamping(p)`. Not used by the notebook. THEORY ONLY

Amplitude damping versus phase damping

	amplitude damping (γ)	**phase damping** (λ)
physical process	energy loss $\lvert 1\rangle \to \lvert 0\rangle$	loss of phase information, no energy exchange
populations	$\rho_{11} \mapsto (1-\gamma)\rho_{11}$	unchanged
coherence	$\times\sqrt{1-\gamma}$	$\times\sqrt{1-\lambda}$
fixed point	$\lvert 0\rangle$ only	every diagonal state
unital?	no	yes
time constant	T_1: $\gamma = 1 - e^{-\Delta t/T_1}$	T_φ: $\sqrt{1-\lambda} = e^{-\Delta t/T_\varphi}$
equivalent Pauli channel?	no	yes: phase flip with $1-2p = \sqrt{1-\lambda}$
`stim`?	no	as a phase flip, yes

Both damp the coherence; only amplitude damping changes populations. Together they give **thermal relaxation** with $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$.

All single-qubit channels on the Bloch sphere ILLUSTRATIVE EXPERIMENT



x–z cross-section of the Bloch ball (dashed) and its image (blue) for $p = \gamma = \lambda = 0.3$. Computed from the Kraus operators and checked with CUDA-Q density-matrix simulations (largest difference 1.1e-16).

- bit flip keeps x; phase flip keeps z; bit-phase flip keeps y (shrinks both x and z); depolarizing shrinks everything by 0.6
- amplitude damping **shifts** the ball towards $|0\rangle$; phase damping squeezes it to the z-axis

Single-qubit noise models: summary

channel	Kraus operators	Bloch map	unital	CUDA-Q
bit flip	$\sqrt{1-p}I,\ \sqrt{p}X$	$(r_x, cr_y, cr_z),\ c = 1-2p$	yes	`BitFlipChannel`
phase flip	$\sqrt{1-p}I,\ \sqrt{p}Z$	(cr_x, cr_y, r_z)	yes	`PhaseFlipChannel`
bit-phase flip	$\sqrt{1-p}I,\ \sqrt{p}Y$	(cr_x, r_y, cr_z)	yes	`YError`
depolarizing	$\sqrt{1-p}I,\ \sqrt{p/3}\{X, Y, Z\}$	$(1-\frac{4p}{3})\vec{r}$	yes	`DepolarizationChannel`
amplitude damping	$\mathrm{diag}(1, \sqrt{1-\gamma}),\ \sqrt{\gamma}\,\lvert 0\rangle\langle 1\rvert$	$(\sqrt{1-\gamma}r_{x,y},\ \gamma+(1-\gamma)r_z)$	no	`AmplitudeDampingChannel`
phase damping	$\mathrm{diag}(1, \sqrt{1-\lambda}),\ \mathrm{diag}(0, \sqrt{\lambda})$	$(\sqrt{1-\lambda}r_{x,y},\ r_z)$	yes	`PhaseDamping`
any channel	user-defined $\{K_k\}$	affine map	—	`KrausChannel`

Used by the notebook: depolarizing, bit flip, phase flip, amplitude damping (plus `Depolarization2`, `Pauli2`, `KrausChannel` on CNOTs).

Two-qubit gate noise

Two-qubit gates are typically several times noisier than single-qubit gates (indicative). Models used by the notebook after every CNOT:

- depolarizing: `Depolarization2(p)` — one joint channel, failure probability p
- bit/phase flip: `Pauli2` with $p(1-p)$ on IX and XI (or IZ, ZI) and p^2 on XX (ZZ) = independent flips
- amplitude damping: `KrausChannel`$\{K_i \otimes K_j\}$ — independent on both qubits

No-error probability of the 15-qubit GHZ circuit (1 H + 14 CNOT): depolarizing $(1-p)^{15}$; bit flip $(1-p)^{1+2\cdot 14} = (1-p)^{29}$.

Numbers

$p = 0.01$: depolarizing $0.99^{15} = 0.8601$; bit flip $0.99^{29} = 0.7472$.
`stim`, 15-qubit GHZ at $p = 0.05$: bit flip gives population 0.2319, parity 0.1996.

Error propagation

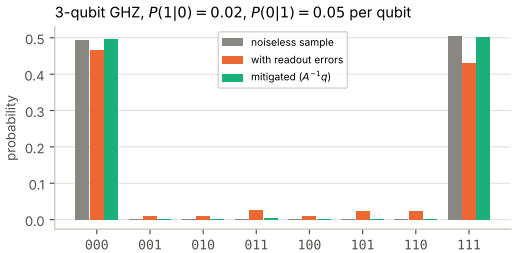
A CNOT copies an X error from control to target and a Z error from target to control: errors spread through entangling layers.

Measurement errors and readout noise ILLUSTRATIVE EXPERIMENT

The measured *bit* is wrong with some probability — a classical error after the quantum measurement:

$$A = \begin{pmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{pmatrix}, \quad \mathbf{q} = A\mathbf{p}.$$

Example: $P(1|0) = 0.02$, $P(0|1) = 0.05$ (asymmetric: $|1\rangle$ may relax during readout). $|+\rangle$ is read as 1 with probability 0.485. A perfect 3-qubit GHZ shows population $\frac{1}{2}(0.98^3 + 0.95^3) \approx 0.899$.



Noiseless CUDA-Q sample, readout errors applied per bit: population 1.0000 → 0.8979; mitigated 0.9987.

Not modelled in `Noise_Simulation.ipynb`: its measurements are ideal.

Thermal relaxation: T_1 and T_2

- T_1 (relaxation): $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$; per step $\gamma = 1 - e^{-\Delta t/T_1}$
- T_2 (coherence): $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$
- $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$, hence $T_2 \leq 2T_1$
- at finite temperature the qubit relaxes to a thermal state, not exactly to $|0\rangle$ (generalized amplitude damping)

Worked examples

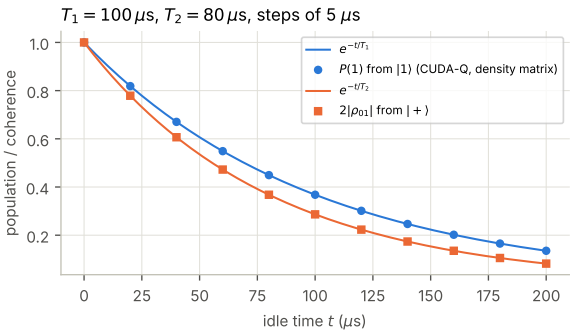
$T_1 = 50\,\mu$s: $P(1) = \frac{1}{2}$ after $T_1 \ln 2 = 34.7\,\mu$s.
$T_1 = 100$, $T_\varphi = 200\,\mu$s $\Rightarrow$ $T_2 = 100\,\mu$s.
$T_2 = 100\,\mu$s, 50 ns gates: $e^{-nt_g/T_2} \geq 0.9$ for $n \leq 210$ gates.

Typical orders of magnitude

superconducting qubits: $T_1, T_2 \sim 10$–$500\,\mu$s, gates 20–500 ns; trapped ions: T_2 seconds or more, gates μs–ms. Ratios coherence/gate time matter, not absolute times. (Indicative literature values, not measured here.)

T_1 and T_2 simulated step by step

ILLUSTRATIVE EXPERIMENT



Each 5 μs idle step applies one `KrausChannel` = amplitude damping γ followed by phase damping λ, with $T_\varphi = 133.3\,\mu$s chosen so that $T_2 = 80\,\mu$s.

- CUDA-Q density matrix agrees with e^{-t/T_1}, e^{-t/T_2} to 8.9e-16
- the identity "idle" gate is `rx(idle)` with `idle = 0.0` passed at run time (see "Pitfalls")

Error rates, gate errors and benchmarking

- **process fidelity** $F_{\text{pro}} = \langle \Phi^+ | J(\mathcal{E}) | \Phi^+ \rangle$ — depolarizing: $1 - p$
- **average gate fidelity** $F_{\text{avg}} = \frac{dF_{\text{pro}}+1}{d+1}$ — depolarizing qubit: $1 - \frac{2p}{3}$
- **error rate** $r = 1 - F_{\text{avg}}$; $p = 0.003 \Rightarrow r = 0.002$
- **randomized benchmarking**: random Clifford sequences of length m; survival $A\alpha^m + B$; $r = \frac{(d-1)(1-\alpha)}{d}$ ($\alpha = 0.99 \Rightarrow r = 0.005$); robust to SPAM errors

Average $\neq$ worst case

A coherent over-rotation and a stochastic error with the same r give very different results after many gates (Part 1). RB reports the average under twirling.

In this repository

RB and process tomography are **THEORY ONLY**; the notebook characterises circuits by population, parity, Hellinger and state fidelity.

Noise channels in circuit execution

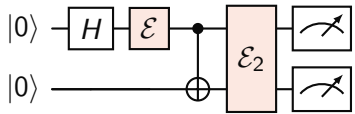
A noisy circuit = ideal gates, each followed by its channel:

$$\rho_{\text{out}} = \mathcal{E}_G \circ \mathcal{U}_G \circ \cdots \circ \mathcal{E}_1 \circ \mathcal{U}_1(\rho_0), \quad \mathcal{U}(\rho) = U\rho U^\dagger.$$

Accumulation: for Pauli channels with error probability p at G locations, the error-free branch has weight $(1-p)^G$, so

$$F \geq (1-p)^G \approx e^{-pG}.$$

Deeper and wider circuits tolerate proportionally less noise.



The bound is not tight

6-qubit GHZ, depolarizing $p = 0.01$: $F = 0.9492 > 0.99^6 = 0.9415$ — e.g. $Z \otimes Z$ after a CNOT leaves GHZ unchanged. For non-Pauli noise (amplitude damping) the notebook does not claim the bound.

Noise injection in CUDA-Q IN REPOSITORY

```
model = cudaq.NoiseModel()
model.add_all_qubit_channel("h",
       cudaq.DepolarizationChannel(p))
model.add_all_qubit_channel("x",
       cudaq.Depolarization2(p), num_controls=1)

# shot-based (stim, trajectories)
counts = cudaq.sample(ghz_z, 15, shots_count=10000,
                      noise_model=model)
# exact expectation value (density matrix)
cudaq.set_target("density-matrix-cpu")
zz = cudaq.observe(kernel, cudaq.spin.z(0),
                   noise_model=model).expectation()
```

- a `NoiseModel` attaches channels to *gate names*; `num_controls=1` selects CNOTs
- the channel acts *after* every matching gate, on the qubits it touches
- `sample` / `observe` take `noise_model=`; `get_state` does not (CUDA-Q 0.16) ⇒ `cudaq.set_noise(model)` ... `unset_noise()`
- only noise-capable targets apply it: `stim`, `density-matrix-cpu`, `nvidia` (trajectories)

Pitfalls found while preparing these slides ILLUSTRATIVE EXPERIMENT

1. Constant identity gates vanish

```
rx(0.0, q[0]) # constant: removed -> no noise
rx(idle, q[0]) # idle = 0.0 passed at run time
```

With `DepolarizationChannel(0.3)` on `rx`: $\langle Z \rangle = 1.0$ vs. 0.6. Idle noise needs a gate that survives compilation.

2. Keep the noise model alive

`cudaq.set_noise(build_noise_model(...))` with a temporary object crashed the process (segmentation fault) in `get_state`: the model was garbage-collected. The notebook keeps a reference — its comment says so.

3. Noiseless targets ignore noise

`qpp-cpu` / `nvidia` without a noise-capable method return ideal results; the notebook refuses `statevector`.

Three ways to simulate noise

method	CUDA-Q target	memory	supports	result
density matrix	`density-matrix-cpu`	$16 \cdot 4^N$ B ($N = 13$: 1.07 GB)	any Kraus channel, any gate	exact ρ
trajectories (Monte Carlo)	`nvidia` + noise model (GPU)	$16 \cdot 2^N$ B per trajectory	any channel	statistical
stabilizer	`stim`	$\sim N^2$ bits	Clifford gates + Pauli noise only	statistical, hundreds of qubits

Trajectories: for each shot, pick Kraus jumps at random (probability $\mathrm{Tr}\, K\rho K^\dagger$), keep a pure state; the average over many trajectories reproduces ρ. Memory ratio to the density matrix at $N = 20$: $2^{20} \approx 10^6$.

Stabilizer: Gottesman–Knill; GHZ with depolarizing noise is fine, rotations by arbitrary angles, T gates or amplitude damping are not. The notebook raises an error instead of silently ignoring such noise.

Shot-based simulation

- `stim` / trajectories: every shot samples one error pattern and one outcome $\Rightarrow$ counts directly
- density matrix: probabilities $=$ diag ρ exactly; the notebook draws `SHOTS` counts with a seeded `multinomial` (statistically identical to measuring)
- why? noisy `cudaq.sample` on `density-matrix-cpu` took ≈ 115 s at 10 qubits (README) vs. ≈ 1 s for `get_state`
- shot noise: $\pm\sqrt{P(1-P)/S}$; at $S = 10^4$ and $P = 0.9$: ± 0.003

Counts $\rightarrow$ figures of merit

X-basis counts {`00:450, 11:400, 01:80, 10:70`}: parity $= (850 - 150)/1000 = 0.70$.

Ideal $P = (0.5, 0, 0, 0.5)$, noisy $Q = (0.45, 0.05, 0.05, 0.45)$: Hellinger $F_H = 0.90$, TVD $= 0.10$.

Noisy versus noiseless execution: what to measure

figure of merit	definition	sensitive to	needs
GHZ population	$P(0\cdots0) + P(1\cdots1)$	bit flips	Z-basis counts
parity	$\langle X^{\otimes N}\rangle = P(\text{even}) - P(\text{odd})$ (X basis)	phase flips (and bit flips)	extra run with Hadamards
Hellinger fidelity	$\left(\sum_x \sqrt{p_x q_x}\right)^2$	any change of the distribution	counts
total variation distance	$\frac{1}{2}\sum_x \lvert p_x - q_x\rvert$	any change	counts
state fidelity	$\langle\psi\lvert\rho\rvert\psi\rangle$	everything, incl. coherences	density matrix
purity	$\mathrm{Tr}\,\rho^2$	mixedness	density matrix
no-error probability	$(1-p)^G$	— (a bound, for Pauli noise)	gate count

All are computed in `Noise_Simulation.ipynb` IN REPOSITORY; the noiseless run ($p = 0$) is always included as a check.

The structure of Noise_Simulation.ipynb IN REPOSITORY

Sec.	Content
1	problem statement, channels, figures of merit
2	imports, versions, GPUs
3	**input parameters** (only cell to edit)
4	validation, gate counts, `build_noise_model`
5	utilities, `auto` method choice, backend
6	expected results, memory, time estimate
7	kernels, drawing, circuit statistics
8	execution with memory monitoring
9	ideal vs. noisy plots and sweep table
10	final benchmark report (JSON + PNG)

Section 3 defaults

`N_QUBITS = 15`, `CIRCUIT = "ghz"` (or `"mirror"`, `MIRROR_LAYERS = 2`)
`NOISE_MODEL = "depolarizing"` (`bit_flip`, `phase_flip`, `amplitude_damping`)
`NOISE_PROBABILITY = 0.01`
`NOISE_SWEEP = [0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1]`
`MEASURE_X_BASIS = True`, `SHOTS = 10000`, `SEED = 42`
`BACKEND = "auto"`, `SIMULATION_METHOD = "auto"`

Exports (`exports/noise/`, not committed): circuit drawing (`.txt`, `.png`), QIR (`.ll`), report (`.json`, `.png`).

`build_noise_model`, line by line

```
def build_noise_model(kind, p):
    model = cudaq.NoiseModel()
    if p == 0.0:
        return model # empty = noiseless
    if kind == "depolarizing":
        one, two = cudaq.DepolarizationChannel(p), cudaq.Depolarization2(p)
    elif kind == "bit_flip":
        probs = [0.0] * 15 # Pauli2 order: IX IY IZ XI XX ...
        probs[0] = probs[3] = p * (1 - p) # flip on one qubit only
        probs[4] = p * p # flip on both qubits
        one, two = cudaq.BitFlipChannel(p), cudaq.Pauli2(probs)
    elif kind == "phase_flip":
        probs = [0.0] * 15
        probs[2] = probs[11] = p * (1 - p) # IZ and ZI
        probs[14] = p * p # ZZ
        one, two = cudaq.PhaseFlipChannel(p), cudaq.Pauli2(probs)
    else: # amplitude damping
        k0 = np.array([[1, 0], [0, np.sqrt(1 - p)]], dtype=complex)
        k1 = np.array([[0, np.sqrt(p)], [0, 0]], dtype=complex)
        one = cudaq.AmplitudeDampingChannel(p)
        two = cudaq.KrausChannel([np.kron(a, b) for a in (k0, k1)
                                                for b in (k0, k1)])
    for gate in ("h", "x", "ry", "rz"):
        model.add_all_qubit_channel(gate, one)
    model.add_all_qubit_channel("x", two, num_controls=1)
    return model
```

- $p = 0$: an empty model, the noiseless reference
- `Pauli2` with $p(1-p), p(1-p), p^2$: exactly two independent single-qubit flips
- amplitude damping on CNOTs: the 4 products $K_i \otimes K_j$ — a valid two-qubit Kraus set
- same p for every gate; no idle noise, no measurement noise, no crosstalk
- checked in the verifier: CUDA-Q's channels equal the Kraus sets of the slides

Kernels and execution, line by line

```
@cudaq.kernel
def ghz_x(n: int):
    qubits = cudaq.qvector(n)
    h(qubits[0])
    for i in range(n - 1):
        x.ctrl(qubits[i], qubits[i + 1])
    h(qubits) # X-basis readout
    mz(qubits)

def run_once(kernels, kind, p, basis):
    model = build_noise_model(kind, p)
    if chosen_method == "density_matrix":
        cudaq.set_noise(model)
        try:
            rho = np.array(cudaq.get_state(kernels[name], *args))
        finally:
            cudaq.unset_noise()
        counts, exact = as_counts_from_rho(rho, N_QUBITS, SHOTS)
        return counts, exact, rho
    result = cudaq.sample(kernels[name], *args,
                          shots_count=SHOTS, noise_model=model)
    return {b: c for b, c in result.items()}, None, None
```

- GHZ in the Z basis (`ghz_z`) and the X basis (`ghz_x`: extra Hadamards, also noisy)
- mirror circuit: $R_Y R_Z$ layers + CNOT ladder, then the inverse — ideally $|0 \cdots 0\rangle$, non-Clifford
- the noise model is *not* drawn: channels are inserted at simulation time
- `try/finally`: the global noise is always removed again
- every p of the sweep runs in both bases ($8 \times 2 = 16$ runs)

Method selection and safeguards IN REPOSITORY

`SIMULATION_METHOD = "auto"`:

1. a GPU is visible → `trajectory` (`nvidia`)
2. CPU, GHZ, Pauli noise → `stabilizer` (`stim`)
3. otherwise → `density_matrix` (`density-matrix-cpu`)

Example: CPU, GHZ, amplitude damping ⇒ density matrix.

Refused on purpose

- `stabilizer` with amplitude damping or the mirror circuit
- `statevector` (ignores noise)
- `BACKEND = "hardware"`: “noise models are a simulator feature”
- density matrix on GPU, trajectories on CPU (not offered by CUDA-Q)

The implementation at a glance

simulator (recorded)	`stim` (CPU), CUDA-Q 0.16.0	shots	10 000, seed 42
other methods	`density-matrix-cpu`, `nvidia` trajectories (GPU, not run)	circuits	GHZ (default), mirror
qubits	15 (density matrix: keep ≤ 10; `stim`: hundreds)	gate locations	1 H + 14 CNOT (GHZ)
noise models	depolarizing, bit flip, phase flip, amplitude damping	error probability	p per gate, sweep $0 \ldots 0.1$
gate-level noise	after `h`, `x`, `ry`, `rz` and every CNOT	measurement noise	none
fidelity measures	population, parity, Hellinger, TVD, state fidelity, purity	bound	$(1-p)^G$ (Pauli noise)
benchmark output	time phases, memory timeline, JSON report	hardware	refused (simulator only)

Recorded versus re-executed results

p	population	$\langle X^{\otimes 15} \rangle$	$(1-p)^G$	outcomes
0	1.0000	1.0000	1.0000	2
0.001	0.9873	0.9660	0.9851	48
0.002	0.9807	0.9324	0.9704	56
0.005	0.9458	0.8374	0.9276	64
0.01	0.8936	0.6910	0.8601	96
0.02	0.8009	0.4834	0.7386	196
0.05	0.5724	0.1518	0.4633	564
0.1	0.3232	0.0160	0.2059	1165

Re-executed with the notebook's model, seed and call order on `stim`: identical to the recorded outputs.

- population $\geq (1-p)^G$ everywhere: the bound holds and is not tight
- the parity falls much faster: phase errors and the extra Hadamards
- outcomes: $2 \rightarrow 1165$ — probability leaks into many bitstrings
- recorded run time 0.25 s (16 runs); re-executed sweep 0.41 s
- the density-matrix path is not part of the recorded outputs; it was run for these slides (next slides)

Bit flips, phase flips and depolarizing noise compared ILLUSTRATIVE EXPERIMENT

noise	p	Z population	X parity
depolarizing	0.01	0.8936	0.6920
depolarizing	0.05	0.5728	0.1522
bit flip	0.01	0.7501	0.7382
bit flip	0.05	0.2319	0.1996
phase flip	0.01	1.0000	0.5580
phase flip	0.05	1.0000	0.0504

15-qubit GHZ, notebook model, `stim`, 10 000 shots.

- phase flips: population exactly 1, parity collapses — invisible in the Z basis
- bit flips: both drop; two failure chances per CNOT make them harsher than depolarizing at the same p
- depolarizing: $\frac{2}{3}$ of single-qubit errors flip Z outcomes, $\frac{2}{3}$ flip phases
- the README's "parity 0.56" for phase flips is reproduced (0.5580 at $p = 0.01$)

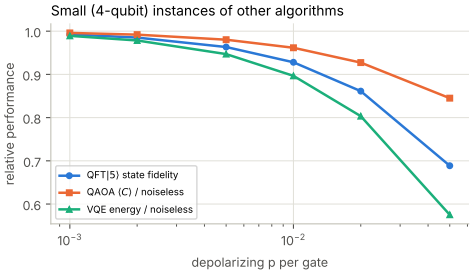
The density-matrix path: fidelity and purity ILLUSTRATIVE EXPERIMENT

noise	p	$F = \langle \mathrm{GHZ}\|\rho\|\mathrm{GHZ}\rangle$	$\mathrm{Tr}\,\rho^2$	population
depolarizing	0.01	0.9492	0.9013	0.9633
depolarizing	0.05	0.7686	0.5959	0.8275
bit flip	0.01	0.9045	0.8191	0.9045
bit flip	0.05	0.6009	0.3749	0.6009
phase flip	0.01	0.9004	0.8206	1.0000
phase flip	0.05	0.6569	0.5492	1.0000
amplitude damping	0.01	0.9495	0.9034	0.9527
amplitude damping	0.05	0.7824	0.6453	0.8106

6-qubit GHZ, notebook model, `set_noise` + `get_state`.
Independent NumPy simulation agrees to 1.3e-15.

- state fidelity and purity need ρ: only this path gives them
- phase flip: population 1 but fidelity 0.66 at $p = 0.05$
- bit flip: fidelity = population (X-type errors never create a relative phase between $|0\cdots0\rangle$ and $|1\cdots1\rangle$)
- amplitude damping: outcomes biased towards $0\cdots0$
- mirror circuit (4 qubits, 32 + 12 noisy gates, $p = 0.01$): success 0.6938 $\geq$ bound 0.6426; noiseless 1.000000

Noise on QFT, QAOA and VQE circuits ILLUSTRATIVE EXPERIMENT



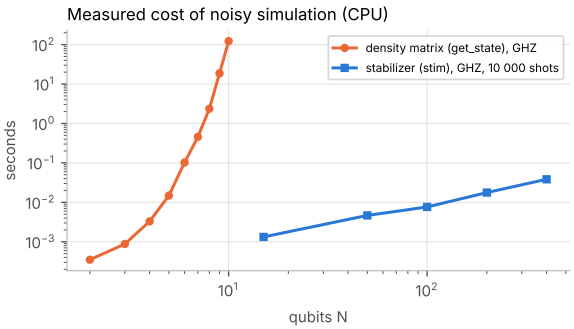
p	QFT state fidelity	QAOA $\langle C \rangle$ (ring, 4)	VQE energy (TFIM, 4)
0	1.0000	3.0000	-3.4141
0.001	0.9926	2.9879	-3.3772
0.002	0.9852	2.9759	-3.3406
0.005	0.9634	2.9408	-3.2332
0.01	0.9280	2.8849	-3.0612
0.02	0.8613	2.7819	-2.7427
0.05	0.6887	2.5350	-1.9638

Small 4-qubit versions on `density-matrix-cpu`, depolarizing p after every gate:

- **QFT** of $|5\rangle$ (H and controlled phases): state fidelity 0.9280 at $p = 0.01$
- **QAOA**, ring of 4, $p = 1$, angles optimised without noise: $\langle C \rangle$ 3.0000 $\rightarrow$ 2.8849
- **VQE**, 4-qubit TFIM, hardware-efficient ansatz (exact ground energy -3.4270, noiseless VQE -3.4141): -3.0612

These are *not* the circuits of `QFT.ipynb`, `QAOA.ipynb`, `VQE.ipynb` (those notebooks model only shot noise). Noisy VQE energies stay above the ground energy (the variational bound still holds for mixed states) but move away from it.

Noise benchmarking: the cost of simulation ILLUSTRATIVE EXPERIMENT



Measured on the 4-core CPU container:

- density matrix, GHZ with depolarizing noise: 10 qubits took 122.46 s for one `get_state` — growing about $\times 7$ per qubit
- `stim`: 400 qubits, 10 000 shots in 0.04 s
- GPU trajectories would sit in between (not measured)

Benchmark noise studies with the cheapest exact method for the circuit class.

Error mitigation, correction and simulation are different things

	noise simulation	error mitigation	error correction
goal	*predict* what noisy hardware does	*estimate* ideal expectation values from noisy runs	*protect* the quantum state
acts on	a classical model	measurement statistics (post-processing)	the state, during the computation
examples	density matrix, trajectories, stim	ZNE, readout mitigation, PEC, symmetry checks	repetition code, surface code
cost	classical memory/time	extra shots (variance grows)	many qubits, fast feedback, below threshold
in this repository	`Noise_Simulation.ipynb` IN REPOSITORY	lecture experiments ILLUSTRATIVE EXPERIMENT	lecture experiment ILLUSTRATIVE EXPERIMENT

Dynamical decoupling is different again: it changes the physical evolution during idle periods.

Zero-noise extrapolation (ZNE)

Measure $E(\lambda)$ at amplified noise $\lambda \geq 1$ and extrapolate to $\lambda = 0$.

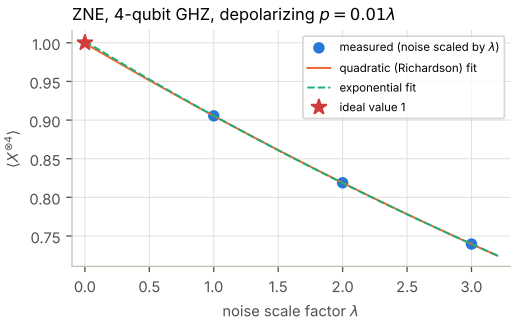
- linear: $E_0 = 2E(1) - E(2)$
- Richardson ($\lambda = 1, 2, 3$): $E_0 = 3E(1) - 3E(2) + E(3)$
- exponential: $E = ae^{-b\lambda}$

Amplifying noise on hardware: gate folding $G \to GG^\dagger G$ ($\lambda = 3$ for that gate; fold 4 of 10 gates: $\lambda = 1.8$), or pulse stretching.

Costs and caveats

- biased: depends on the model ($0.80, 0.64 \to$ linear 0.96, exponential 1.00)
- variance amplified by $\sqrt{\sum \gamma_k^2}$ ($\sqrt{19} \approx 4.4$ for Richardson)
- only expectation values, not states; no protection of long computations

ZNE on a GHZ parity ILLUSTRATIVE EXPERIMENT



4-qubit GHZ, $\langle X^{\otimes 4} \rangle$, notebook depolarizing model at $p = 0.01\lambda$ (exact density matrix):

λ	1	2	3
value	0.9055	0.8189	0.7396

linear 0.9921, Richardson 0.9994, exponential 1.0021 (> 1: unphysical overshoot). Here p is scaled in software; on hardware the noise must be amplified physically.

Readout-error mitigation ILLUSTRATIVE EXPERIMENT

1. calibrate: prepare each basis state, measure, estimate A (columns = prepared states)
2. measure the experiment: frequencies $\mathbf{q}$
3. solve $A\mathbf{p} = \mathbf{q}$ (least squares with $\mathbf{p} \geq 0$ in practice)

Worked example

$P(0|0) = 0.98$, $P(0|1) = 0.05$, measured $q_0 = 0.60$:
$0.93 p_0 = 0.55 \Rightarrow p_0 = 0.591$.

3-qubit GHZ (figure on the readout slide): population 0.8979 $\rightarrow$ 0.9987.

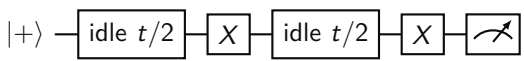
Limits

negative quasi-probabilities (minimum here -0.0009); shot noise amplified; $2^N \times 2^N$ matrix $\Rightarrow$ tensor-product or correlated sparse models for many qubits; does nothing for gate errors.

Dynamical decoupling

An idle qubit with an unknown but *static* detuning δ acquires the phase δt. A π pulse in the middle reverses the sign of the phase accumulated afterwards (Hahn echo):

$$R_Z(\tfrac{\varphi}{2})\,X\,R_Z(\tfrac{\varphi}{2}) = X \quad (XR_Z(a)X = R_Z(-a)).$$



Sequences (CPMG, XY4, . . .) repeat this to suppress noise slower than the pulse spacing.

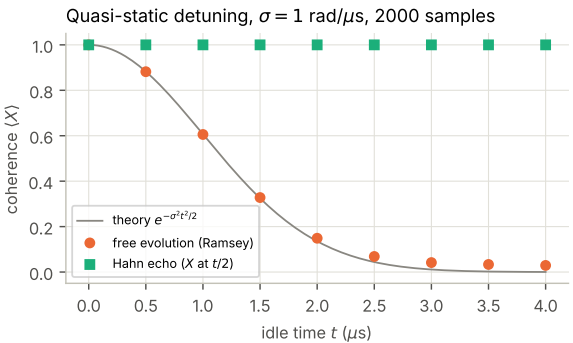
Quasi-static detuning, $\sigma = 1$ rad/μs

Free: $\overline{\langle X\rangle} = e^{-\sigma^2 t^2/2}$, at $t = 2\,\mu\text{s} = 0.135$.
Echo: 1 for every static δ.

Not for white noise

Independent phase flips each step ($p = 0.02$, 20 steps): $\langle X\rangle = 0.96^{20} = 0.4420$ with *and* without echo (CUDA-Q: 0.4420 / 0.4420).

Echo versus free evolution ILLUSTRATIVE EXPERIMENT

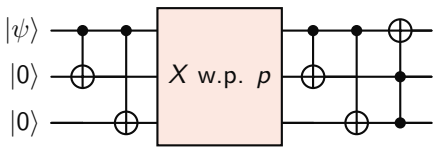


CUDA-Q (qpp-cpu): average of $\langle X \rangle$ over 2000 random detunings.

- free evolution follows $e^{-\sigma^2 t^2/2}$ (at $t = 2\,\mu$s: 0.149; theory 0.135; Monte-Carlo error $\approx \pm 0.02$)
- the echo restores $\langle X \rangle = 1.000$ exactly
- real noise has a spectrum: DD filters the slow part

Quantum error correction fundamentals

3-qubit bit-flip code:
$\alpha\,|0\rangle + \beta\,|1\rangle \mapsto \alpha\,|000\rangle + \beta\,|111\rangle$; measure the parities Z_0Z_1, Z_1Z_2 (syndrome); flip the indicated qubit.



encode noise decode + majority vote (Toffoli)

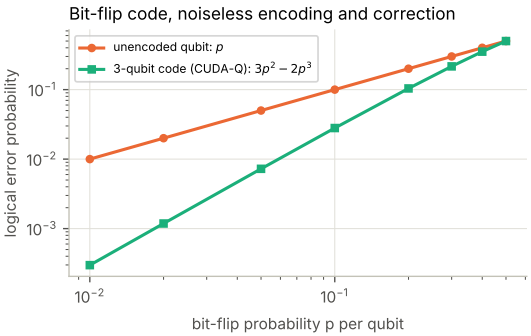
Logical error $P_L = 3p^2 - 2p^3$:

- $p = 0.1$: $P_L = 0.028$; $p = 0.01$: $P_L = 0.0003$
- better than a bare qubit iff $p < \frac{1}{2}$
- protects only against X; Shor's 9-qubit code and surface codes protect against all Pauli errors

Surface codes: $p_L \approx A(p/p_{\text{th}})^{(d+1)/2}$, threshold $p_{\text{th}} \sim 1\,\%$; $d = 3 \to 5$ at $p/p_{\text{th}} = 0.1$ gains a factor 10.

THEORY ONLY

The bit-flip code, simulated ILLUSTRATIVE EXPERIMENT



CUDA-Q density matrix: bit-flip channel on identity gates (`rx(idle)`) after encoding; noiseless decoding.

- matches $3p^2 - 2p^3$ to 2.6e-16
- $p = 0.01$: 0.000298; $p = 0.1$: 0.0280
- idealised: perfect gates and syndrome extraction. With noisy gates a fault-tolerant design and a threshold are needed.

Twirling: turning coherent errors into Pauli noise THEORY ONLY

Insert a random Pauli P before a gate G and the compensating $P' = GPG^\dagger$ after it (for Clifford G, P' is again a Pauli): the ideal circuit is unchanged. Averaging over random compilations,

$$\mathcal{E}_{\text{twirled}}(\rho) = \tfrac{1}{4} \sum_P P^\dagger \mathcal{E}(P\rho P^\dagger) P$$

is a **Pauli channel** with the same average fidelity.

Why it matters

- coherent errors stop adding in amplitude (Part 1: 0.995 vs. 0.036)
- makes Pauli noise models — like the notebook's — realistic
- basis of probabilistic error cancellation and of RB's interpretation

Simulated noise versus hardware noise

What simple noise models miss

- **correlated** and **crosstalk** errors between neighbouring qubits
- **non-Markovian** noise (memory, $1/f$ spectra), drift over hours
- **leakage** out of the qubit subspace
- idle-qubit errors while others are being gated (the notebook adds noise only after gates)
- readout errors, state-preparation errors
- device-specific connectivity (extra SWAPs)

Good practice

calibrate a model from device data (T_1, T_2, RB/IRB rates, readout matrices), validate it against small hardware experiments, and report simulation and hardware results separately.

In this lecture

every number is a simulation on a CPU; nothing was run on a quantum processor.

Hardware noise considerations and future directions

- gate errors around 10^{-3} (two-qubit) and below on leading platforms; QEC below threshold demonstrated on small surface codes
- noise-aware compilation: mapping, scheduling, DD insertion
- learning noise models (Pauli-noise learning, cycle benchmarking)
- scalable simulation: GPU trajectories, tensor networks, stabilizer + few non-Clifford gates
- from mitigation to logical qubits

Natural extensions of `Noise_Simulation.ipynb`

- readout-error channel and its mitigation
- idle noise from T_1, T_2 and gate durations
- noise models for the other notebooks' circuits (QFT, QAOA, VQE)
- GPU trajectory runs at 20–30 qubits; ZNE

Try it yourself: `Noise_Simulation.ipynb`

Edit only Section 3:

- `NOISE_MODEL = "phase_flip"`: population stays 1, parity collapses
- `N_QUBITS = 100`: `stim` still runs in seconds; fidelity at fixed p falls faster
- `SIMULATION_METHOD = "density_matrix"`, `N_QUBITS = 8`: state fidelity and purity
- `NOISE_MODEL = "amplitude_damping"`: outcomes biased towards 0
- `CIRCUIT = "mirror"`, `N_QUBITS = 6`: a non-Clifford benchmark

Reproduce these slides:

- `cd presentations/noise_source`
- `python make_figures.py` (re-executes the notebook's sweep and all experiments, $\sim$ 5 min)
- `latexmk -pdf noise_lecture.tex`

Test yourself: `assessments/Noise_Question_Paper.pdf` (80 questions) and the separate Answer Book.

Exercises

1. Find the Bloch vector and purity of $\rho = \begin{pmatrix} 0.6 & 0.2i \\ -0.2i & 0.4 \end{pmatrix}$.
2. Apply amplitude damping with $\gamma = 0.5$ to $|1\rangle$ and to $|+\rangle$.
3. Show that the phase-flip channel with p equals phase damping with $\lambda = 4p(1-p)$.
4. A depolarizing channel (CUDA-Q) with $p = 0.06$ acts twice. Give the single equivalent p.
5. $T_1 = 120\,\mu$s, $T_2 = 90\,\mu$s. Find T_φ and the coherence of $|+\rangle$ after $30\,\mu$s.
6. No-error probability of the notebook's 20-qubit GHZ circuit with depolarizing $p = 0.005$.
7. ZNE gives $E(1) = 0.70$, $E(2) = 0.52$, $E(3) = 0.40$. Linear and Richardson estimates?
8. 3-qubit bit-flip code at $p = 0.05$: logical error?

Answers

1. $r_z = 0.2$, $\rho_{10} = -0.2i = \frac{r_x + i r_y}{2} \Rightarrow r_y = -0.4$: $\vec{r} = (0, -0.4, 0.2)$; purity $\frac{1+0.2}{2} = 0.6$.
2. $|1\rangle \mapsto \mathrm{diag}(0.5, 0.5)$; $|+\rangle \mapsto \left(\begin{smallmatrix} 0.75 & 0.354 \\ 0.354 & 0.25 \end{smallmatrix}\right)$ ($\frac{\sqrt{0.5}}{2} = 0.354$).
3. Coherence factors: $1 - 2p = \sqrt{1-\lambda} \Rightarrow \lambda = 1 - (1-2p)^2 = 4p(1-p)$.
4. $(1 - 0.08)^2 = 0.8464 = 1 - \frac{4p}{3} \Rightarrow p = 0.1152$.
5. $\frac{1}{T_\varphi} = \frac{1}{90} - \frac{1}{240} = 0.006944 \Rightarrow T_\varphi = 144\,\mu\mathrm{s}$; $|\rho_{01}| = \frac{1}{2} e^{-30/90} = 0.358$.
6. $G = 1 + 19 = 20$: $0.995^{20} = 0.905$.
7. Linear $2(0.70) - 0.52 = 0.88$; Richardson $3(0.70) - 3(0.52) + 0.40 = 0.94$.
8. $3(0.05)^2 - 2(0.05)^3 = 0.0075 - 0.00025 = 0.00725$.

Summary

- Noise turns pure states into mixed states: density matrices, Bloch ball, purity, fidelity.
- Channels: $\mathcal{E}(\rho) = \sum K \rho K^\dagger$, $\sum K^\dagger K = I$ (CPTP); Kraus sets are not unique.
- Bit/phase/bit-phase flip and depolarizing shrink the ball; amplitude damping also shifts it (T_1); phase damping $\leftrightarrow T_\varphi$; $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$.
- Coherent errors accumulate quadratically, stochastic ones linearly; readout errors are classical.
- CUDA-Q: `NoiseModel` + `stim` / density matrix / trajectories; the notebook's GHZ sweep reproduced exactly.
- Mitigation (ZNE, readout, DD) estimates or suppresses; only QEC protects the state; simulation predicts.

Further reading

- M. Nielsen, I. Chuang, *Quantum Computation and Quantum Information*, Ch. 8 and 10
- J. Preskill, *Lecture notes for Physics 219*, Ch. 3
- K. Temme, S. Bravyi, J. Gambetta, *Error mitigation for short-depth circuits*, PRL 119 (2017)
- P. Krantz et al., *A quantum engineer's guide to superconducting qubits*, APR 6 (2019)
- NVIDIA CUDA-Q documentation: noise modelling
- This repository: `Noise_Simulation.ipynb`, `assessments/Noise_*.pdf`

Syllabus coverage: topics 1–40

#	topic	slides
1	Introduction to ideal quantum computing	4
2	Ideal versus noisy circuits	5, 39
3	Sources of quantum noise	6
4	Decoherence	7
5	Quantum state fidelity	8
6	Gates and imperfect operations	9
7	Bit-flip errors	14
8	Phase-flip errors	15
9	Bit-phase-flip errors	16
10	Depolarizing noise	17
11	Amplitude damping	18
12	Phase damping	19
13	Amplitude vs phase damping	20
14	Quantum channels	12
15	Kraus operators	12, 13
16	Density matrices	10
17	Pure and mixed states	11
18	CPTP maps	13
19	Single-qubit noise models	22, 21
20	Two-qubit gate noise	23

#	topic	slides
21	Measurement errors	24
22	Readout noise	24, 47
23	Thermal relaxation	25, 26
24	T_1 and T_2 times	25, 26
25	Noise channels and circuit execution	28
26	Noise injection in simulation	29, 30
27	Shot-based simulation	32
28	Noisy vs noiseless execution	33, 40
29	State and process fidelity	8, 27, 41
30	Error rates and gate errors	27
31	Noise simulation using CUDA-Q	29, 31
32	Repository walkthrough	34, 35, 36, 37, 38
33	Noise on QFT, VQE, QAOA	42
34	Noise benchmarking	39, 43
35	Error mitigation	44
36	Zero-noise extrapolation	45, 46
37	Readout-error mitigation	47
38	Dynamical decoupling	48, 49
39	Error-correction fundamentals	50, 51
40	Hardware noise, future directions	53, 54